Raspberry Pi 5 Dual Wi-Fi Optimization Guide

Optimized for: 10 units, high-density video streaming (Internal) & low-req Internet (USB Dongle)

1. Hardware Assignment Overview

To ensure maximum stability for 20+ streaming devices, we prioritize the internal Pi chip for heavy local traffic:

- **wlan0 (Internal Pi Wi-Fi):** Dedicated to **Controllab** (High-bandwidth video).
- **wlan1 (Edimax USB Dongle):** Dedicated to **Free-TAU** (Low-demand Internet).

2. Implementation Steps

Run these commands on your master Pi to configure the interfaces.

Configure Internal Video Network (wlan0)

Assign the internal Wi-Fi to your video network. We set a low metric for high priority but disable its use as an internet gateway:

Bash
```
sudo nmcli connection modify "Controllab" connection.interface-name wlan0
ipv4.route-metric 100 ipv4.never-default yes
```

Configure Internet-Only Network (wlan1)

Assign the USB dongle to the internet network. It will act as the default gateway for the web but will have lower priority for local traffic:

Bash
```
sudo nmcli connection modify "Free-TAU" connection.interface-name wlan1
ipv4.route-metric 400 ipv4.never-default no
```

3. Verify Routing Strategy

Apply the changes:

Bash
```
sudo nmcli connection down "Free-TAU" && sudo nmcli connection up "Free-TAU"
sudo nmcli connection down "Controllab" && sudo nmcli connection up
"Controllab"
```

Run `ip route` to confirm the logic:

- **Default via wlan1:** All internet traffic leaves through the Edimax dongle.

- **Internal Subnet via wlan0:** Video traffic stays on the high-performance internal chip (Metric 100).

4. Preparation for Mass Cloning (10 Units)

Before creating your SD card image, you must generalize the connection files located in `/etc/NetworkManager/system-connections/`:

1. **Remove MAC Addresses:** Open each `.nmconnection` file and delete the `mac-address=` line. This prevents the image from failing when the hardware ID changes on the next Pi.
2. **Lock Interface Names:** Ensure the `interface-name=` field is set to `wlan0` or `wlan1` respectively to maintain your performance strategy.